Characterization and prediction of students' population: Universidad Autónoma de Occidente - case of study

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Highlights

- We found out that people in stratum 4 is 35% more likely to complete the enrollment process than the ones in stratum 3.
- 18% of students come from Valle del Cauca department, without taking Cali into account.
- 71% of confidence in the association between men registered in engineering and their enrollment, without any kind of aid.

Background

In the last decade the university has aimed to guarantee admission to higher education of the largest possible population, and they procure to be more assertive with the training and scholarships offered to students. That is why, they require having a profile analysis of demand for new students, in order to provide access aid to the students that are more likely to enroll. Additionally, the institution needs to understand better the tendencies and projections in the amount of potential students and adapt its business model in response to this behavior.

The Solution

We developed a solution that allows the university to be cost-effective in the reach out process.

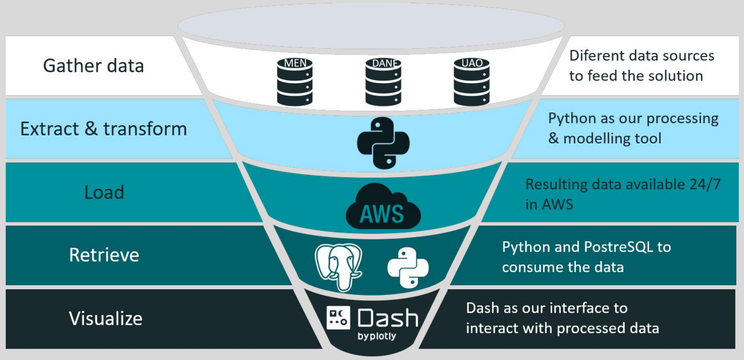
TWO MAIN FEATURES

A CHARACTERIZATION APPROACH: Allows the university to know **who** may be interested in enroll.

A PREDICTION APPROACH: Allow them to know **who** to focust its efforts on to complete the enrollment.

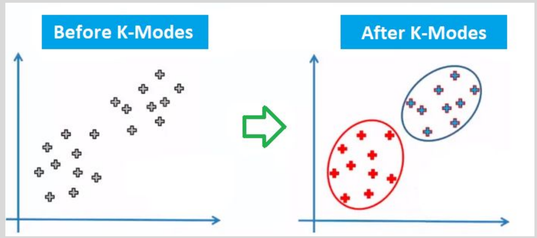
LEAD TO A MORE **COST-EFFECTIVE** PROCESS

The infrastructure of the developed solution was implemented as follows.



Clustering

The K-Modes algorithm is a technique designed in order to segment individuals, using their attributes as input, in order to generate groups of people whose characteristics are similar, in turn allowing a clear difference between the groups. The elbow and Silhouette methods are used as decision criteria to find the optimal number of groups. With this approach, two groups were found.



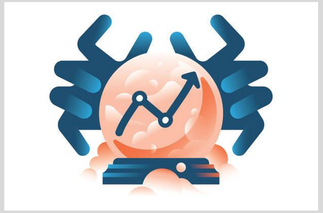
Characterization



The method seeks to find frequent and relevant patterns between characteristics in individuals and express them in the form of conditional rules. The confidence, support and lift indicators are used to filter the most relevant rules.

Prediction Model

Logistic regression is a multivariate statistical technique, which uses a set of predictive variables of the individual and returns the probability that this individual enrolls or not in university, Thus allowing to generate strategies that help these people to enroll in university.



Clustering results



Segment 1 is mainly characterized by having people belonging to the engineering faculty in 63%. On the other hand, this cluster is given mainly by people of the male gender (72%) and belonging to socioeconomic strata 3 and 4 (37% and 20% respectively).



Segment 2 is characterized by containing people who are mainly in the faculty of administrative sciences (54%). Represented mainly by the female gender with 70% of the total of the people belonging to this cluster. This cluster is represented by people belonging to socioeconomic strata 2 and 3.

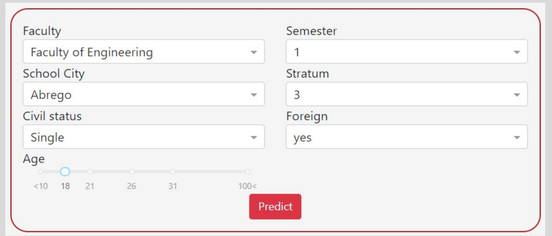
Characterization results



With the characterization methodology, rules were found such as:

1. Socio-economic stratum and financial aid affect the decision-making of registration.
2. A lot of men decide to take engineering faculty majors and a lot of women decide to take majors from the administration faculty.

Prediction results



By entering certain features of a person, the model gives the probability that the person will enroll or not in a specific major.

